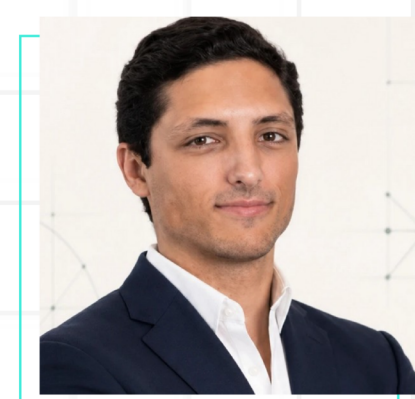


From Design to Digital Twin: Physics and AI for Fatigue-Aware Floating Offshore Wind Towers

From Knowledge to Competitiveness

Portugal-US Partnerships 2026 Summit

MIT Portugal Program



João Alves Ribeiro^{1,2,3}

jpar@mit.edu

Supervisors: Francisco Pimenta¹, Sérgio M. O. Tavares², Faez Ahmed³

Co-author: Bruno Alves Ribeiro⁴

- 1. Faculty of Engineering, University of Porto (LAETA-INEGI, CONSTRUCT)
- 2. TEMA, University of Aveiro
- 3. Massachusetts Institute of Technology (DeCoDE Lab)
- 4. TU Delft and Brown University

under the Doctoral Grant SFRH/BD/151364/2021 and MIT Portugal Seed Project 2024.14532.MIT Energy

THE CHALLENGE

Floating wind is key for Portugal's deep Atlantic waters [1].

- Optimised towers are critical for floating wind
- 25-year fatigue life to prove
- Thousands of simulations per design
- Loads unmeasured once installed

OUR APPROACH

The same structural physics governs design, simulation, learning and monitoring.

One family of physics + AI tools on the IEA 22 MW floating reference turbine.

1 DESIGN FLOAT [2]

Redesign the tower for a 25-year fatigue life.

3 SIMULATE LESS NeuroFLOAT [4]

Simulate a physics-selected subset, let AI fill the rest.

2 BENCHMARK FLOATBench [3]

Test whether AI surrogates hold outside their training data.

4 MONITOR FLOATSense [5]

Estimate tower loads at 11 heights from tower-top sensors.

1 FLOAT [2]

IEA reference

~9 months

FLOAT tower 25-year design life

0 25 yr

25 years

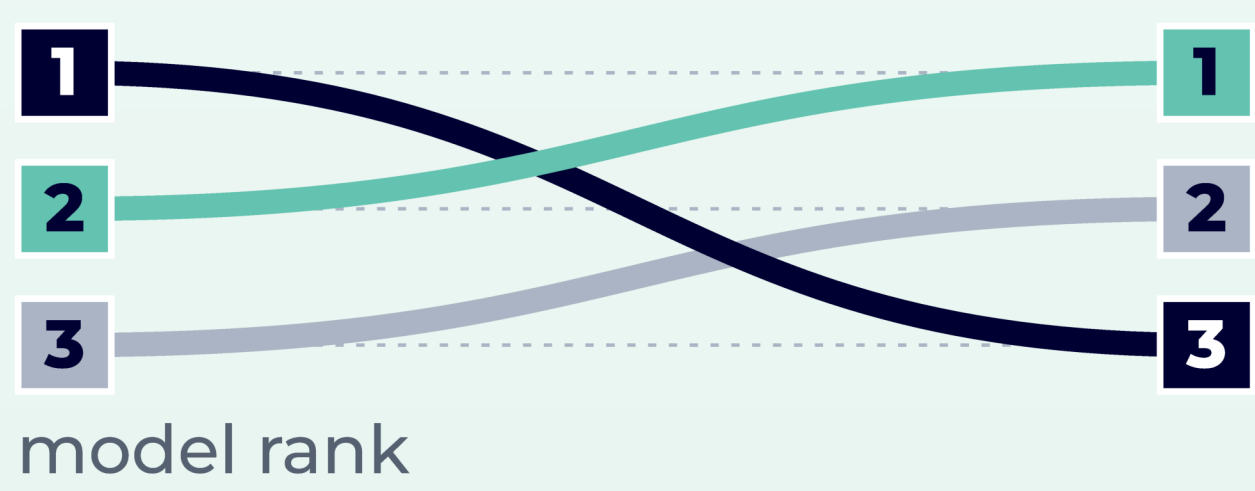
tower fatigue life (from ~9 months)

Merged into the official IEA 22 MW model (NREL/DTU).

2 FLOATBench [3]

Random

Regime-aware

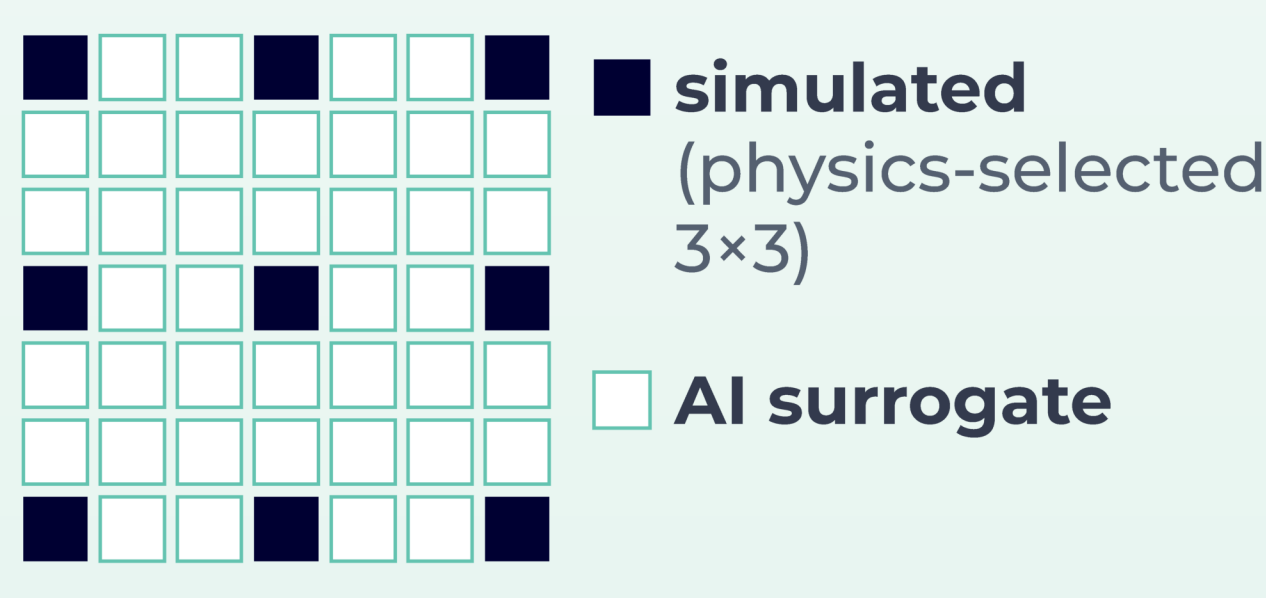


582,120

damage-labelled samples, 3 towers

Model rankings can invert when extrapolating.

3 NeuroFLOAT [4]

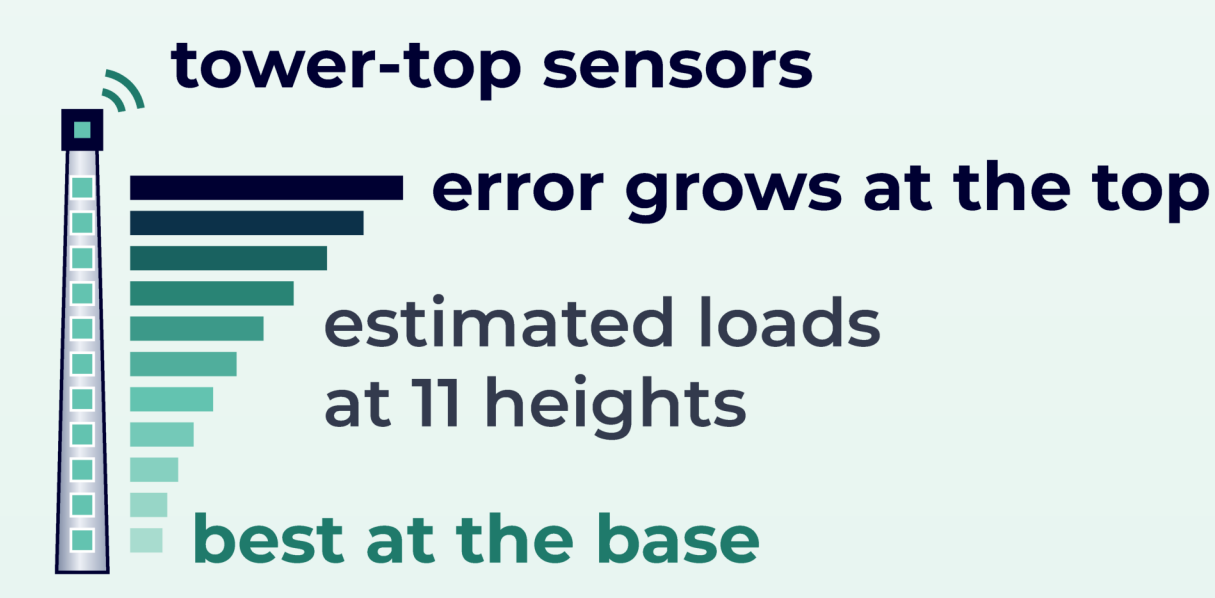


5.4x

fewer simulations (6,468 to 1,188)

Lifetime damage within ~0.5% of the full campaign.

4 FLOATSense [5]



11 heights

loads from tower-top sensors

Best model at the base can lose accuracy at the top.

FOR INDUSTRY

- Before installation: far fewer simulations.
- After installation: loads from tower-top sensors. (simulation-validated; field next)

PORTUGAL-US PARTNERSHIP

PORTUGAL
University of Porto (FEUP) | University of Aveiro
UNITED STATES
MIT | Brown University
INTERNATIONAL
TU Delft | IEA Wind

KEY MESSAGE

Physics provides reliability, AI provides scale.



ALL PROJECTS

References [1] Alves Ribeiro et al., Applied Energy 397 (2025) 126294. [2] FLOAT, Ocean Engineering 362 (2026) 126285. [3] FLOATBench, [4] NeuroFLOAT, [5] FLOATSense, all under review. MIT Portugal Seed Project: <https://doi.org/10.54499/2024.14532.MIT>.

Funded by:

